

Smart Attendance System

Software Engineering Project

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1. Introduction

The **Smart Attendance System** is a web-based application designed to automate daily class attendance using dynamic QR codes and GPS location checks.

Instead of taking roll call manually, the teacher displays a QR code on the screen, and students scan it using their mobile phones.

2. Background

In most colleges and universities, attendance is still taken manually using paper registers or by calling out roll numbers one by one. This traditional approach has several basic problems:

- **Time Wasted:** Calling out names takes 5 to 10 minutes out of every lecture.
- **Proxy Attendance:** Students frequently sign the register or mark attendance for their absent friends.
- **Manual Errors:** Teachers have to manually count and calculate attendance percentages at the end of the month, which leads to mistakes.
- **Record Damage:** Paper registers can easily get lost, damaged, or misplaced over time.

To solve these issues, we need a digital system that automates the process and makes it harder for students to mark fake attendance.

Digital attendance systems aim to reduce these problems by recording attendance electronically instead of on paper. Once attendance is stored digitally, it can be viewed, searched, and summarized far more easily than a paper register, and there is no risk of the physical sheet being lost.

QR codes are a convenient way to support digital attendance because they can be generated instantly, scanned quickly using an ordinary smartphone camera, and changed frequently. In this project, a new QR code is generated for every attendance session and is refreshed automatically after a short period of time, which makes it harder for a code to be copied and reused after the class has ended.

Location authentication adds a second layer of verification on top of the QR code. When a student scans the QR code, the system also checks the student's current device location against the location recorded for the class session. This gives the teacher more confidence that the student was actually near the classroom at the time of scanning, rather than marking attendance from somewhere else using a shared QR image.

Finally, a centralized database is used to store all attendance records, along with student, teacher, and course information, in one place. This makes it possible for teachers to view

attendance for their courses at any time, and for students to check their own attendance history and percentage, without depending on a paper register maintained by someone else.

3. Problem Statement / Need of the Project

3.1 Problems with the Traditional Attendance Process

- Manual attendance takes up classroom time that could be used for teaching.
- Teachers have to maintain attendance records manually for every class and every course.
- Manual records can contain human errors, such as marking the wrong student.
- Maintaining attendance records over a full semester becomes difficult to organize.
- Calculating attendance percentages by hand for each student is inconvenient and slow.
- Paper-based attendance sheets can be lost, misplaced, or damaged.
- Traditional attendance methods, such as calling out names or passing a sheet, can allow proxy attendance, where one student marks attendance on behalf of another.
- It is difficult for a teacher to verify whether a student marking attendance is actually present at the attendance location.
- Teachers do not have an immediate digital view of who is present or absent during a session.
- Students are not able to easily check their own attendance percentage or history without asking the teacher or checking a physical register.

3.2 Need for the Smart Attendance System

The problems described above show that there is a clear need for a digital system that can record attendance more efficiently and with fewer manual steps. The Smart Attendance System is proposed to address this need.

The proposed system combines four checks before attendance is recorded for a student: a temporary QR code that identifies the active session, location authentication that compares the student's device location with the session location, verification that the student is actually enrolled in the course, and a duplicate-attendance check that prevents the same student from marking attendance twice for one session.

Together, these checks are intended to make the attendance process more organized and more difficult to misuse than a simple paper register or a roll call. It is important to note that this combination improves the reliability of the attendance process, but it does not make proxy attendance impossible in every situation, since it still depends on factors such as device location

accuracy and permissions granted by the student's device, as explained later in the Limitations section.

4. Objectives

The objectives of the Smart Attendance System are:

1. To digitize the attendance process for classroom sessions.
2. To reduce the time required to take attendance manually.
3. To reduce human errors that occur in manually recorded attendance.
4. To provide separate dashboards for the Admin, Teacher, and Student, each showing features relevant to that user.
5. To generate temporary QR codes for each attendance session.
6. To allow students to scan the QR code using their device camera.
7. To verify the student's device location before marking attendance.
8. To verify that the student is enrolled in the relevant course before marking attendance.
9. To prevent duplicate attendance from being recorded for the same student in the same session.
10. To automatically store attendance records in a database.
11. To allow teachers to view attendance records for their courses.
12. To allow students to view their attendance history and attendance percentage.
13. To allow administrators to manage users, courses, and system records.
14. To provide a simple and user-friendly attendance management system suitable for a classroom setting.

5. Users / Stakeholders

The final system has three main users: Admin, Teacher, and Student. Each user has a separate login and a separate set of features suited to their role.

5.1 USER 1 _Admin

The Admin is responsible for managing the overall system. The Admin does not take or mark attendance directly, but manages the accounts, courses, and enrollment information that the Teacher and Student features depend on.

Admin features include:

- Admin login
- Manage students (add, update, delete student records)
- Manage teachers (add, update, delete teacher records)

- Manage courses (add, update, delete courses)
- Assign students to courses
- Manage course enrollment
- View attendance records
- Manage basic system information

5.2 User 2 — Teacher

The Teacher uses the system to start and manage attendance sessions for the courses assigned to them, and to review attendance afterward.

Teacher features include:

- Teacher login
- Access to the teacher dashboard
- View assigned courses
- Select a course before starting a session
- Start an attendance session
- Generate a temporary QR code for the session
- Set the attendance location using the device's current location
- View live attendance while the session is open
- Close an attendance session
- Review attendance records for their courses

The QR code generated for a session is temporary and expires after a short period, such as 60 seconds. While the session remains open, the system generates a new code automatically, so the code shown on screen keeps changing during the attendance session.

5.3 User 3 — Student

The Student uses the system to mark their own attendance by scanning the QR code shown by the Teacher, and to review their attendance history afterward.

Student features include:

- Student login
- Access to the student dashboard
- View enrolled courses
- Open the QR scanner
- Use the device camera to scan the Teacher's QR code
- Grant location access when prompted

- Send location information for verification
- System verifies whether the student is within the allowed attendance area
- System verifies QR/session validity
- System verifies course enrollment
- System checks whether attendance has already been marked for the session
- Attendance is marked only when all required conditions are satisfied
- View attendance history
- View attendance percentage

6. Location Authentication

Location authentication is an important part of the final system. It works as an additional check on top of the QR code, so that attendance depends on both scanning the correct code and being close to the classroom.

The process works as follows:

1. The Teacher starts an attendance session for a course.
2. The system generates a temporary QR code associated with that session and also records the Teacher's current location as the session location.
3. The Student opens the scanner and scans the QR code using the device camera.
4. The system obtains the Student's current location after the Student grants location permission.
5. The system compares the Student's location with the session location recorded in step 2.
6. If the Student is within the defined allowed radius and all other checks (session validity, enrollment, duplicate check) pass, attendance is marked.
7. If the Student is outside the allowed radius, attendance is rejected and the Student is informed that they are too far from the class location.

NOTE: Location authentication provides an additional layer of verification, but it cannot guarantee a student's physical presence with absolute certainty. GPS and device location accuracy can vary depending on the device and surroundings, and location permission must be granted by the student for this check to work at all. This limitation is realistic and is described further in the Limitations section of this document.

7. QR Code Attendance Process

The complete attendance process, from the Teacher starting a session to the Student's attendance being stored, follows these steps:

1. Teacher logs in.
2. Teacher selects a course.
3. Teacher starts an attendance session.
4. System creates a temporary session and an associated token.
5. QR code is generated for the session.
6. Student logs in.
7. Student opens the camera-based QR scanner.
8. Student scans the QR code.
9. System verifies the QR code and session validity.
10. System verifies that the student is enrolled in the course.
11. System verifies the student's location against the session location.
12. System checks for duplicate attendance in this session.
13. Attendance is stored in the database.
14. Student receives an on-screen confirmation.
15. Teacher can view the updated attendance list.

8. User-wise Requirements / Features

This section lists the detailed requirements for each user of the system in tabular form, using a requirement ID, the requirement name, and a short description of what the requirement means.

8.1 User 1: Admin Requirements

ID	Requirement	Description
AR-1	Admin Login	The Admin logs in using a username and password before accessing any system feature.
AR-2	Manage Students	The Admin can add, update, and delete student records, including name and basic login details.
AR-3	Manage Teachers	The Admin can add, update, and delete teacher records, including name and basic login details.
AR-4	Manage Courses	The Admin can add, update, and delete course records such as course name and course code.
AR-5	Assign Students to Courses	The Admin can enroll a student in one or more courses so that the student can mark attendance for those courses.
AR-6	Manage Course Enrollment	The Admin can update which students are enrolled in a given course, including removing a student from a course if required.

ID	Requirement	Description
AR-7	View Attendance Records	The Admin can view attendance records for any course, teacher, or student stored in the system.
AR-8	Manage System Information	The Admin can view and manage basic system information, such as the list of registered users and courses.

8.2 User 2: Teacher Requirements

ID	Requirement	Description
TR-1	Teacher Login	The Teacher logs in using a username and password to access the Teacher dashboard.
TR-2	View Assigned Courses	The Teacher can view the list of courses assigned to them by the Admin.
TR-3	Select a Course	The Teacher selects a specific course before starting an attendance session for that course.
TR-4	Start Attendance Session	The Teacher starts an attendance session for the selected course at the beginning of the class.
TR-5	Generate Temporary QR Code	The system generates a temporary QR code linked to the attendance session. The QR code expires after a short period, such as 60 seconds, and is regenerated automatically while the session remains open.
TR-6	Set Attendance Location	The system records the Teacher's current location (the classroom location) when the session is started. This location is later used to verify each student's position.
TR-7	View Live Attendance	The Teacher can view the list of students who have marked attendance while the session is still active.
TR-8	Close Attendance Session	The Teacher can end the attendance session manually, or the session closes automatically once its time limit is reached.
TR-9	View Attendance Records	The Teacher can view attendance records for their own courses, for a single session or over a range of dates.

8.3 User 3: Student Requirements

ID	Requirement	Description
SR-1	Student Login	The Student logs in using a username and password to access the Student dashboard.
SR-2	View Enrolled Courses	The Student can view the list of courses they are currently enrolled in.
SR-3	Open QR Scanner	The Student opens the built-in QR scanner from the dashboard when an attendance session is open.
SR-4	Scan QR Code Using Camera	The Student uses the device camera to scan the QR code displayed by the Teacher.
SR-5	Grant Location Permission	The Student allows the system to access the device's current location through the browser's location permission.
SR-6	Location Verification	The system compares the Student's current location with the Teacher's session location and checks whether the Student is within the allowed radius.
SR-7	Session / QR Validity Check	The system checks that the scanned QR code belongs to an active, non-expired session before proceeding further.
SR-8	Enrollment Verification	The system checks that the Student is enrolled in the course linked to the session before marking attendance.
SR-9	Duplicate Attendance Check	The system checks whether the Student has already marked attendance for this session and rejects a repeated attempt.
SR-10	View Attendance History	The Student can view a list of their past attendance records for each enrolled course.
SR-11	View Attendance Percentage	The Student can view their calculated attendance percentage for each enrolled course.

9. Technology Used

9.1 Frontend

The user interface (dashboards for Admin, Teacher, and Student, along with the QR scanner screen) is built using HTML for structure, CSS for styling, and JavaScript for interactive behavior such as opening the camera and reading the student's location in the browser.

9.2 Backend

The backend of the system is built using Python with the Flask framework. Flask handles user login, session management, QR code generation, and the logic for verifying location, enrollment, and duplicate attendance before a record is saved.

9.3 Database

SQLite is used as the database for this project. It stores user accounts (Admin, Teacher, Student), course information, enrollment records, and attendance records. SQLite is suitable for this project because it is lightweight, requires no separate server, and is enough to handle the scale of data expected in a semester-long academic project.

9.4 Attendance Technology

- QR codes, generated for each attendance session and refreshed automatically while the session is open.
- The device camera, used by the student to scan the QR code.
- Browser/device location services, used to obtain the student's current location for location authentication.

9.5 Development Environment

- Visual Studio Code, used as the code editor for both the backend and frontend code.
- Windows 11, used as the operating system for development and testing.

10. System Overview

At a high level, the system is organized into the following layers:

Admin / Teacher / Student → Frontend (HTML, CSS, JavaScript) → Flask Backend →
Authentication & Validation → SQLite Database

The Admin, Teacher, and Student are the three users who interact with the system through the frontend, which is what they see and use in the browser. The frontend sends requests to the Flask backend, for example to log in, start a session, or submit a scanned QR code along with the student's location.

The Flask backend contains the authentication and validation logic. This is where the system checks the username and password at login, checks whether a QR/session is valid and not expired, checks whether the student is enrolled in the course, checks the student's location against the session location, and checks whether attendance has already been marked for that session.

Once all required checks are passed, the backend saves the relevant information, such as a new attendance record, a new user, or a new course, into the SQLite database. The database is also used to read back information whenever a dashboard needs to display records, such as a teacher viewing attendance or a student viewing their attendance history.

11. Important System Rules

- One student cannot mark attendance twice for the same session.
- Expired QR codes or expired sessions are not accepted by the system.
- A student who is not enrolled in the course is not allowed to mark attendance for that course's session.
- A student outside the allowed attendance location is not allowed to mark attendance.
- All successful attendance records are stored in the database.
- Teachers can view attendance records for the courses assigned to them.
- Students can view their own attendance records, but not the records of other students.
- The Admin can manage user accounts, courses, and enrollment records.

12. Limitations

- The student needs a compatible device with a working camera to scan the QR code.
- Camera permission must be granted by the student for QR scanning to work.
- Location permission must be granted by the student for location authentication to work.
- Poor GPS or location accuracy on some devices may affect the reliability of location verification.
- Depending on how the system is deployed, an internet or local network connection may be required for the student's device to reach the backend.
- QR-based and location-based verification improve the reliability of attendance, but they cannot guarantee a student's physical presence with absolute certainty.
- This project is an academic system built for a 3rd-semester Software Engineering course, and it is not intended to be a production-level commercial attendance platform.